# The Vector Is the Answer?
# One-Pass Residual Readouts vs Fair Loop Scoring

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**Abstract**

On closed-set tasks whose answer is representable in a single forward pass, a small supervised head on a frozen residual stream is a competitive *classifier* relative to the strongest closed-set use of the autoregressive loop. We fit heads to the final-token residual of six open-weight models (Qwen3-0.6B/4B/8B, Mistral-7B, Granite-3.1-8B, DeepSeek-V4-Flash) and compare them to a fairly-conditioned loop — full context, balanced few-shot, scored by next-token log-probability over the same answer set — matched on information access and scoring protocol, and budget-stress-tested at 0.6B (Ext 13; $k \leq 64$ at $4\times$ pad). We do *not* compare against free-form or multi-step generation. On BoolQ (full validation), RuleTaker n2k (*pilot*), and ARC-Challenge (full test), the one-pass readout matches or beats that closed-set scorer at nearly every scale: it wins significantly where the verbal channel is weakest (RuleTaker $+12.0$pt on Granite-8B, $+5.2$pt at 0.6B; BoolQ $+3.0$pt at 0.6B against the budget-max loop), is a *statistical* tie at mid-scale BoolQ (numerical losses of $\sim$0.7–1.1pt, all non-significant), and loses in three ordered regimes — small models on parametric knowledge (ARC), a frontier MoE on serial deduction (RuleTaker pilot at DeepSeek-V4), and Mistral-7B on BoolQ. The map survives label-shuffle nulls, random-projection selectivity, per-length deconfounds, and paired McNemar on identical rows. Sweeping the readout tap across depth, the verdict is assembled by mid-depth: the best mid-layer tap beats the final layer at 9 of 18 cells (6/18 after Benjamini–Hochberg) and never loses significantly. Operationally, on this task class the residual supports a classifier that matches the loop's closed-set scorer — **the vector already has the answer** — with the boundaries of that claim mapped rather than erased.

## 1 Introduction

Autoregressive decoding is the universal serving interface for language models, but it is only one readout of the computation. Every generated token is produced by applying a fixed, next-token-trained unembedding to a residual vector; the answer a model *would* give is thus a particular linear read of a vector that exists, in full, after a single forward pass — and our own linear probes sit within a few points of the MLP everywhere (§3). This paper asks a sharp, falsifiable question about that vector: **on tasks whose answer can be represented in one forward pass, does a small readout head fit to the frozen residual recover the answer — and does the autoregressive loop add anything such a readout lacks?**

The question is easy to answer sloppily. A probe that beats a zero-shot prompt has shown little: zero-shot understates the loop, and a token-probability decision rule is a known-weak baseline [8]. A probe compared only to the model's own self-report has shown something else entirely — that beliefs are present, not that a readout can *replace* the loop at producing the right answer. And any probe result without label-shuffle nulls, random-projection selectivity, and significance testing is one confound away from collapse. Our contribution is therefore as much a **discipline of controls** as a result: we run the comparison the probing literature has stopped short of — frozen open weights, a readout fit to gold labels, against a fairly-conditioned closed-set scoring loop, matched on information and scoring, and budget-stress-tested at small scale (Ext 13).

**Concurrent work.** The observation that activations carry content the response does not report is now independently published: Hazenoot et al. [20] show a linear probe on frozen activations beats the same model's own prompted yes/no answer on ESG concept measurement (11/12 comparisons), with the winning layer mid-network. Their loop baseline is zero-shot prompting — which we show understates the loop (§3, Ext 13) — and the comparison carries no paired significance or selectivity controls. We take the convergence as evidence the premise is ripe; the measurement we run here is the part that was missing.

**Findings.** On BoolQ (full validation, 3270 rows), RuleTaker (n2k pilot), and ARC-Challenge (full test), across six models from 0.6B to a frontier MoE:

1. **Parity or better almost everywhere.** The one-pass readout matches or beats the fair few-shot closed-set scorer at nearly every scale — winning significantly where the loop is weakest, and a statistical tie at mid-scale BoolQ (§3, §4).

2. **The loop wins in three bounded regimes:** small models on parametric knowledge (ARC-Challenge at $\leq$7B scale), a frontier MoE on formal serial deduction (RuleTaker at DeepSeek-V4), and Mistral-7B on BoolQ. The first two gaps close or invert with scale and task type (§5).

3. **The answer is assembled by mid-depth.** Sweeping the readout tap across the residual stream, the best mid-layer tap beats the final layer significantly at 9/18 cells (6/18 after Benjamini–Hochberg) and never loses — the final layer is a *lower bound* on what the residual readout can do (§4.4).

4. **The margins can be large.** Where the loop's verbal channel is weakest, the readout's lead is largest: $+12.0$pt on RuleTaker with Granite-3.1-8B (readout 0.805 vs loop 0.685, McNemar $p = 6.5\text{e}{-12}$).

*Claim discipline.* The readout is at parity or better in most cells; the loop wins in three *regimes* — a frontier MoE on serial deduction (RuleTaker pilot at DeepSeek-V4, $-3.8$pt), Mistral-7B on BoolQ ($-2.2$pt), and small models on parametric knowledge (ARC, $-10.6 \rightarrow -0.7$pt as models grow). The remaining BoolQ cells are statistical ties — four of six sit within $\pm1.2$pt, all non-significant. The contribution is the *map* of where each readout wins, not a blanket verdict — and the map is mostly one color.

**Why it matters.** Later work on residual heads, latent taps, and efficient inference can treat "the answer is already in the vector" as established on a defined task class — not as a serving system we ship here, and not as a claim about all tasks. We state the boundaries explicitly (§5, §6): fine-tuned specialist encoders beat our readout; serial-depth tasks may require the loop; the frozen continuous-latent loop drifts. Within those bounds, the footnote is earned.

**Scope.** A companion position paper explores a serving architecture of kilobyte-scale readout heads (Paper 2). This paper contains no architecture claims and no production system.

# 2 Setup and Method

## 2.1 Two readouts of the same vector

**One-pass readout.** Let $h_T \in \mathbb{R}^d$ be the final-token residual of a frozen model after one forward pass over the task prompt (question + passage, truncated at pad_max $= 384$). We fit a head $f(h_T) \rightarrow \hat{y}$: either multinomial logistic regression or a one-hidden-layer MLP (width 128, 4-seed vote), both on standardized features. The head sees only task labels; the backbone is never updated.

**Fair loop.** The autoregressive baseline is *not* greedy decode — it is the strongest *classifier* the loop provides. The model receives the full context (balanced few-shot exemplars, loop_pad_max 2048–8192) and is scored by next-token log-probability over the closed answer set (Yes/No, or A–D). We report loop.0 (zero-shot) and loop.$k$ ($k$-shot); where noted we sweep $k$ up to 64 with a $4\times$ pad budget (Ext 13). *Scope:* we evaluate the strongest *classifier* the loop offers, not a chain-of-thought loop that generates intermediate tokens before answering. Generated-then-answered chains are a different readout, break the matched closed-set scoring protocol, and — on a serial task such as RuleTaker — are not one-pass-representable by construction; they are out of scope for the matched comparison.

### 2.2 Matching requirements

A readout-vs-loop comparison is only meaningful if neither side is handicapped. We match on three axes: **information access** (both see the same question and passage; the loop additionally sees $k$ exemplars — an advantage we grant it), **scoring protocol** (both produce a distribution over the same closed answer set; accuracy on identical rows), and **supervision budget, stress-tested not fully equalized** (Ext 13 sweeps the loop's exemplar budget to $k \leq 64$ at $4\times$ pad on BoolQ at 0.6B/4B/8B; the head still sees the full gold training split, so we do not claim equal learning — only that extra in-context supervision does not erase the pattern where it was tested).

### 2.3 Controls

Every headline number carries four controls:

- **Label-shuffle null**: the head refit on permuted labels must sit at chance, ruling out leakage and spurious separability.
- **Random-projection selectivity**: a head on random Gaussian projections of the residual. If it retains accuracy, the signal is *diffuse* — no privileged subspace — which is what we find (permuted-basis $\approx$ max-basis $\gg$ noise floor) at every scale.
- **Final-token vs full-context**: a mean-pooled full-context readout must underperform the final-token readout, confirming the verdict is *computed into* the final state rather than read off the passage surface.
- **Paired significance**: McNemar's exact test and paired bootstrap on per-row predictions from identical val rows, not pooled-accuracy eyeballing.
- **Per-length deconfound**: headline cells are re-scored within question- and passage-length terciles; accuracy is flat across length (0.89–0.90 on BoolQ at DeepSeek-V4), so the readout's edge is not an artifact of the loop truncating long rows.

### 2.4 Models and artifacts

Six open-weight models spanning three families and a width ladder: Qwen3-0.6B ($28\times1024$), Qwen3-4B ($36\times2560$), Qwen3-8B ($36\times4096$), Mistral-7B ($32\times4096$), Granite-3.1-8B ($40\times4096$), DeepSeek-V4-Flash ($43\times4096$, FP8 MoE). All runs are batch 8 on model-pinned GPUs (0.6B/4B H200, 8B/Mistral/Granite B200, DSV4 B300). Trained heads are persisted per layer as versioned artifacts ($\sim$0.5–8.4 MB each); per-row predictions persist for paired tests. Run manifests embed the batch tag in cache keys after the Ext 15 cache-identity incident.

## 3 BoolQ: one-pass readout vs the fair loop

BoolQ [10] is binary reading comprehension with the gold answer grounded in a passage — the cleanest case of a one-pass-representable task. We load the standard Hugging Face release (`google/boolq`) and use the full validation split (9427 train / 3270 val).

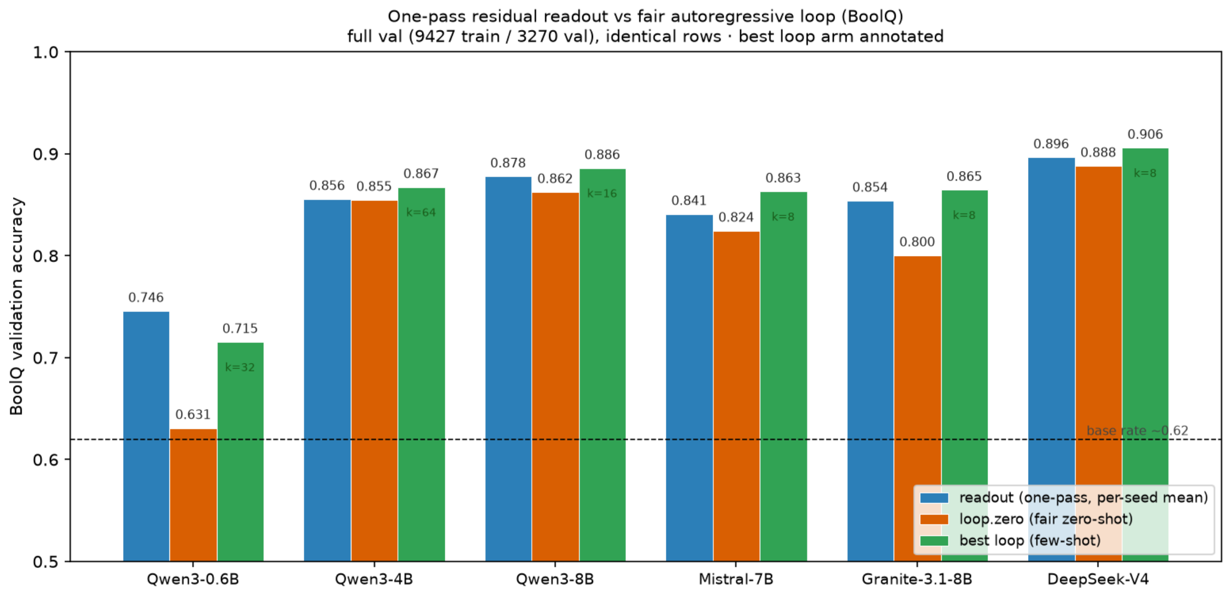


**Figure 1: BoolQ: readout vs loop** across six models (full val).

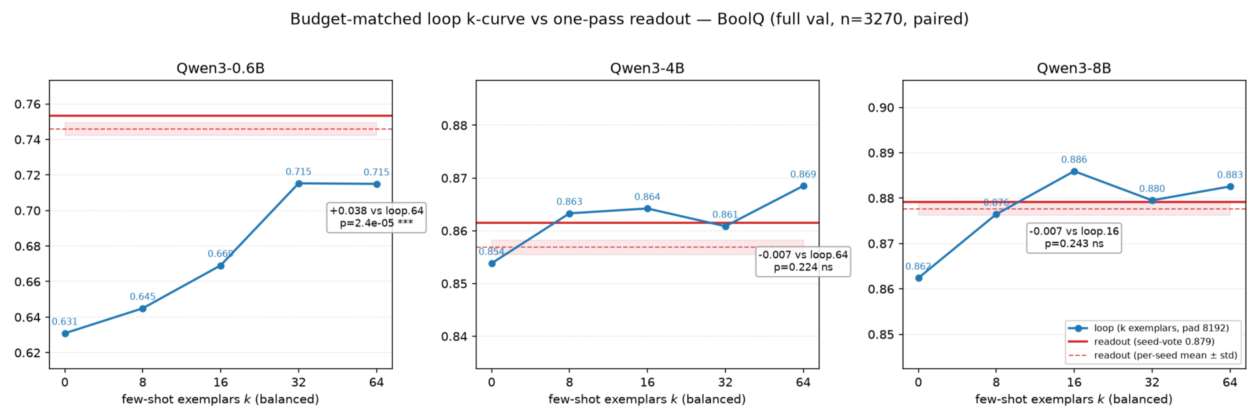


**Figure 2: Supervision budget (Ext 13).** The loop's $k$-curve plateaus by $k$=32 and never passes the readout.

## 3.1 Headline

Across the Qwen and Granite scales, the one-pass readout **wins at 0.6B, is a statistical tie at 4B/8B (numerical losses ∼0.8–1.2pt, ns), and trails the few-shot loop by ∼1pt (ns) at Granite-8B and the frontier MoE** (Table 1). The one exception is Mistral-7B, where the loop wins by 2.2pt ($p = 2.6\text{e}{-03}$) — the single significant loop win on BoolQ, which we flag as a single-model exception (§6).

## 3.2 Supervision asymmetry is bounded (Ext 13)

The natural objection: the head sees thousands of labels, the loop sees $k$ exemplars. We gave the loop its budget back — $k$ swept to 64 balanced exemplars at $4\times$ pad (8192) — at 0.6B/4B/8B. The loop's $k$-curve **plateaus by** $k$=32 (loop.64 − loop.32 = $-0.0003$, $p = 1.00$ at 0.6B) and **never passes the readout where the readout was winning**: at 0.6B the readout's $+3.0$pt over the best loop stays significant ($p = 3.2\text{e}{-05}$); at 4B/8B all cells are exact ties. Supervision starvation is not the explanation (Fig. 2).

### 3.3 Readings

1. **Parity or better at five of six models; the edge is largest where the model is weakest.** The readout's only significant win is at 0.6B — the loop's verbal channel is the bottleneck there, and the readout channel saturates before it. Mistral-7B is the exception (loop $+2.2$pt).

2. **The signal is distributed, not localized.** Random-projection heads retain accuracy (perm $\approx$ max $\gg$ noise) at every scale; no privileged subspace carries the verdict (Table 4).

3. **The verdict concentrates in the final token.** Mean-pooled full-context readouts underperform the final-token readout everywhere (full-context $\ll$ final-token) — the answer is computed into the final state, not read off the passage.

4. **The readout is essentially linear.** A multinomial logistic readout tracks the MLP within a few points at every scale (BoolQ 0.6B linear.max 0.750 vs MLP 0.747; 8B 0.874 vs 0.881), so the verdict is a near-*linear* read of the residual, not a fragile nonlinear probe. At 0.6B, where the residual does not yet hold the answer (ARC $\approx$0.49, near chance), no head can invent it — the readout reads what is there.

*Context length.* The readout reads at pad_max $= 384$ against the loop's 2048–8192. Because the BoolQ gold answer is grounded in the passage, left-truncation could in principle drop long passages asymmetrically; we do not log per-row truncation rates and flag this as a limitation (§7), noting the per-length deconfound above shows no length-dependent accuracy collapse.

### 3.4 FLOPs, precisely

Readout and *scoring* loop are both a single context forward pass — FLOP-equal up to context length. The readout's structural lever is against a *decoding* loop — and here the non-self-termination is **measured**, not assumed: a greedy decode from the "Answer:" prompt never stops (25/25 hit the 300-token cap at 0.6B; 7/8 hit the 1000-token cap at DeepSeek-V4). The parametric win is therefore real and unconditional: **a head instead of a decode loop — no new tokens, no KV-cache.** (We quantify the economics and their boundary against early-exit serving in Table 3; the decode row is measured.)

## 4 RuleTaker and ARC: serial depth and parametric knowledge

BoolQ is passage-grounded. We test the two ways a task can be harder for a one-pass readout: **serial deduction depth** (RuleTaker) and **parametric knowledge with no passage** (ARC-Challenge).

### 4.1 RuleTaker n2k (pilot): matched loop rows, per-depth strata

RuleTaker [12] is closed-world rule reasoning with labeled deduction depth (0/1/2/3/5 + NatLang), loaded via Hugging Face `tasksource/ruletaker`. Protocol: 2000 train / 1000 val (n2k pilot subsample of the full release), loop on the same 1000 rows, $k \in \{0, 8\}$, batch 8. On Qwen, the matched one-pass MLP **beats the fair 8-shot loop at every scale** ($+2.9$ to $+5.2$pt; significant at 0.6B and 4B, $p = 3.8\text{e}{-03}$ and $5.3\text{e}{-03}$; 8B is a near-tie $p = 0.056$). The cross-family models sharpen the picture: Mistral-7B $+7.2$pt ($p = 8.4\text{e}{-04}$), **Granite-3.1-8B $+12.0$pt ($p = 6.5\text{e}{-12}$) — the largest readout margin in the campaign.** The one clean loop win in this pilot is DeepSeek-V4 ($-3.8$pt, $p = 8.5\text{e}{-03}$), driven by the shallow-depth strata.

**Per-depth** (same rows): no sharp "only the loop works past depth $D$" cliff. Accuracy degrades gradually with depth; $d$=5 remains well above chance. The DSV4 loop win concentrates at shallow depths ($d0/d1$, $+10$–15pt); at 0.6B the loop *collapses* with depth ($d5$ loop.8 0.392 $\ll$ readout 0.544). Thin bins ($n = 21$–51) are reported with $n$ and not over-read (Fig. 3).

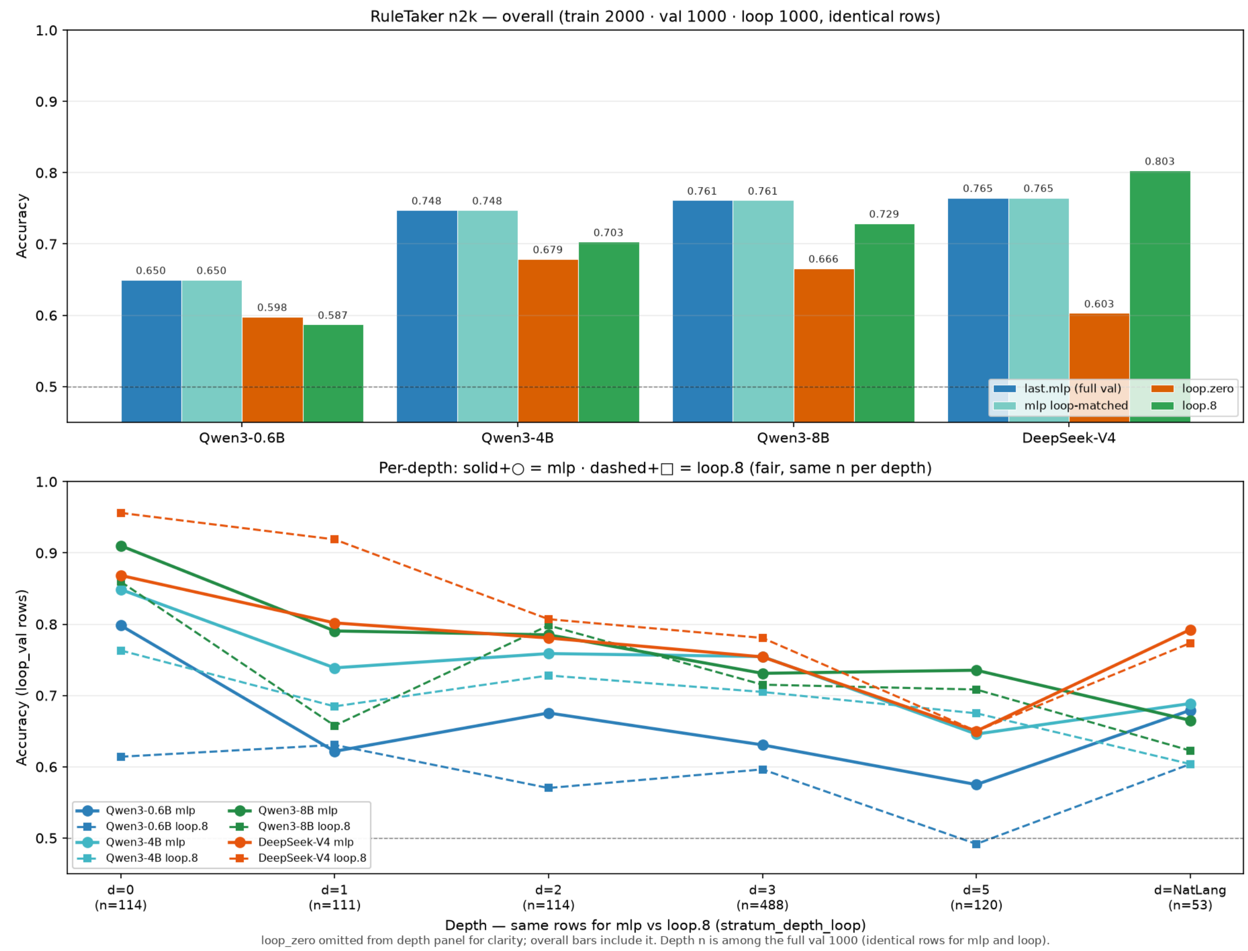


**Figure 3: RuleTaker overall + per-depth strata**, readout vs loop.

### 4.2 ARC-Challenge: the loop's best case

ARC-Challenge [11] is the hard partition of the AI2 Reasoning Challenge (not Chollet's Abstraction & Reasoning Corpus): grade-school science multiple-choice questions that retrieval and co-occurrence baselines miss. We load `allenai/ai2_arc` config `ARC-Challenge` and keep 4-choice A–D items only ($\sim$99% of the set). Answers are parametric science knowledge with no passage — they must come from the weights, not the context. Here the loop wins at small scale: $-10.6$**pt at 0.6B,** $-4.6$**pt at 4B** (both $p < 1\text{e}{-05}$), narrowing to $-0.5$pt at 8B and $-0.7$pt at DSV4 (both ns). The cross-family models replicate the pattern: Mistral-7B $-5.1$pt ($p = 9.3\text{e}{-05}$), Granite-8B $-8.2$pt ($p = 5.8\text{e}{-10}$). The gap **closes monotonically with scale** — few-shot exemplars teach a task format the residual head must otherwise learn from labels, and the advantage shrinks as the parametric answer saturates the residual.

### 4.3 The task-type regularity

Across 18 task$\times$model cells, a single regularity organizes everything (Fig. 4): **the loop wins in three regimes — small models on parametric knowledge (ARC), a frontier MoE on formal serial deduction (RuleTaker at DSV4), and Mistral-7B on BoolQ. Everywhere else the one-pass readout wins or ties.** The ARC crossover tracks *capability*, not family (Granite/Mistral sit where Qwen did at equal accuracy); the RuleTaker crossover tracks *serial structure at frontier scale*; the Mistral BoolQ cell is a single-model exception we flag

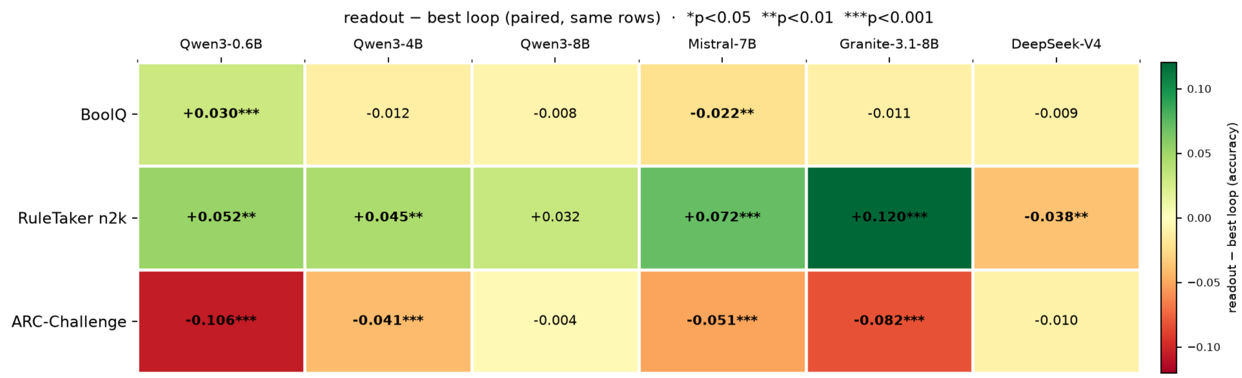


**Figure 4: The task-type regularity.** readout − best loop (paired, same rows) per task×model cell, with McNemar significance. Green everywhere except three red cells — small models on parametric ARC, the frontier MoE on serial RuleTaker, and Mistral-7B on BoolQ.

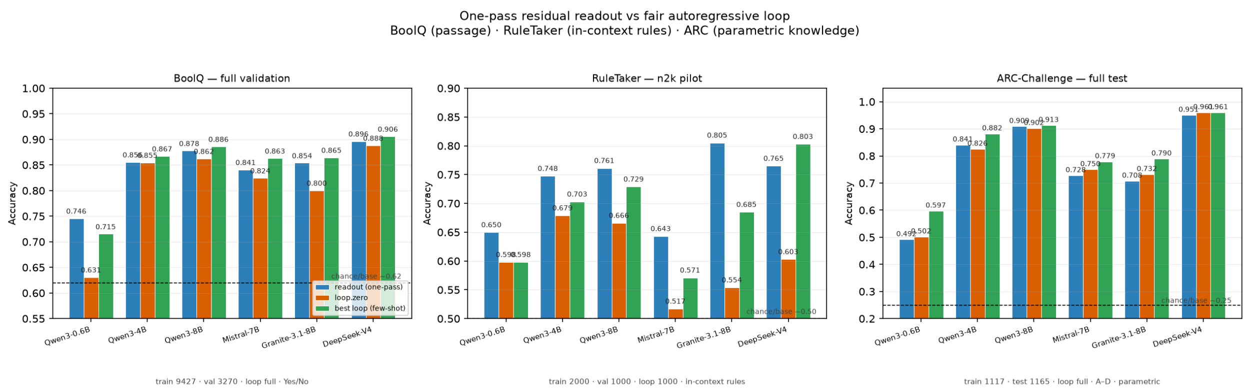


**Figure 5: Three-task head-to-head** (BoolQ · RuleTaker · ARC) across models.

rather than explain (§6).

## 4.4 Readout placement: the final layer is a lower bound

Every headline number above taps the final residual layer. We swept the tap across depth (10 layers, paired McNemar vs the final-layer readout, trained heads persisted at every tap). The curve is the same everywhere: monotone rise, mid-depth plateau, slight terminal decline (Table 2, Fig. 6).

- **The final layer never wins.** Mid-depth taps beat the final layer significantly at **9/18 cells and never lose significantly** — the headline numbers were a lower bound on the readout, not its ceiling. Across the 18 cells, 6 survive a Benjamini–Hochberg FDR correction at $q < 0.05$ (the three borderline cells — BoolQ 4B, RuleTaker 0.6B, RuleTaker 8B — drop out); the "never loses" direction is unchanged.
- **The optimum is task-stable, not scale-driven.** BoolQ/RuleTaker peak at 50–86% depth at every scale; RuleTaker's optimum sits at ∼2/3 depth from 0.6B through DSV4. ARC drifts to the final layer at ≥8B — once the parametric answer saturates the residual, late-layer specialization stops costing anything.
- **Placement converts parity into wins.** Where the loop catches the final-layer readout (BoolQ 8B: L35 ties loop), the best mid-layer tap re-establishes the lead (L30 +1.1pt, $p = 0.016$; on Granite, L20 +1.8pt, $p = 1\text{e}{-03}$).

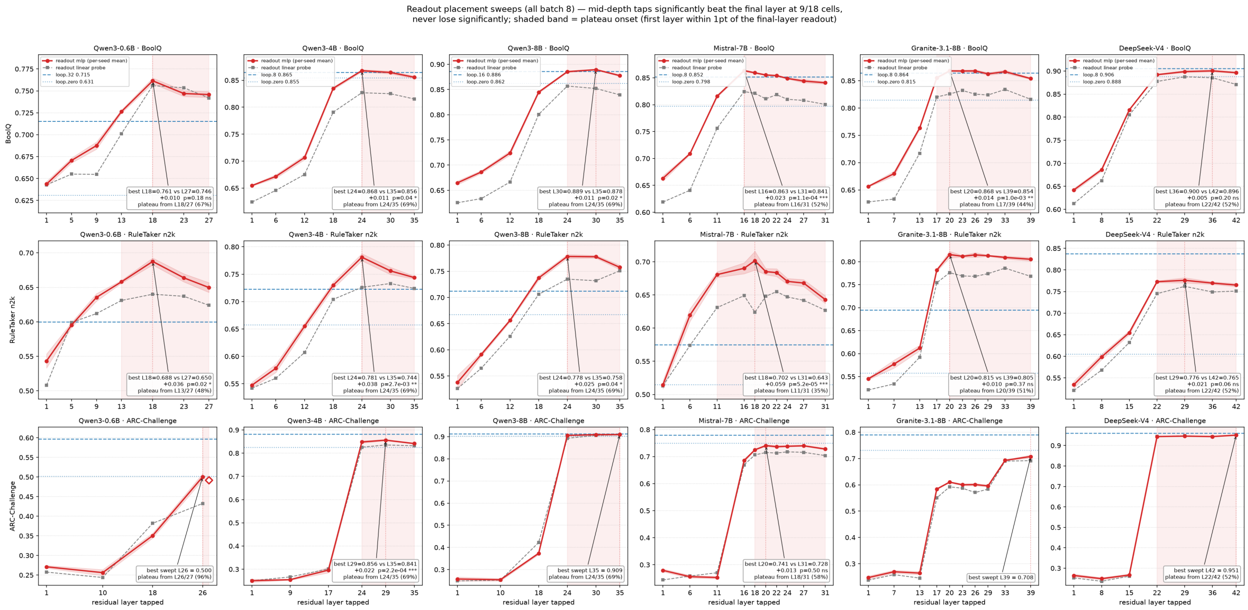


**Figure 6: Readout probe-placement sweeps** (18 cells). One-pass readout accuracy vs tapped residual layer; loop arms as horizontal references; paired McNemar annotation per panel. Mid-depth taps win significantly at 9/18 cells and never lose.

- **Wide-shallow plateaus earlier.** Mistral (32×4096) onsets at 35–58% depth vs Qwen3-8B's uniform 69% at the same width — at fixed width, the shallower stack assembles the verdict sooner. Granite (40×4096, deep) sits between the two.

*Causal hedge.* The monotone-rise → plateau → terminal-decline shape is *consistent with* late-layer specialization toward next-token surface form, but we do not demonstrate it: we never ablated deep layers against generation, and never probed next-token structure per layer. The direct test (a per-layer next-token probe) is future work; "the verdict readout needs less than full depth" and "the final third is load-bearing for decoding" are different claims, and only the first is on the table here.

# 5 The boundary: where the loop could win, and why it doesn't cleanly here

A single forward pass is a bounded-depth computation ($TC^0$ in the circuit- complexity framing); the loop's advantage should appear exactly where serial depth is irreducible. We hunted that regime and found it only weakly exhibitable on frozen base weights:

- **RuleTaker per-depth** shows a gentle slope, not a regime switch — the loop edges the readout by a few points at $d$=5 (noisy thin bins), and the one decisive loop win (DSV4) concentrates at *shallow* depths, where the frontier model's in-context deduction is strong.
- **ARC** shows the loop's real advantage is *parametric retrieval under a learned task format*, not serial compute — and it closes with scale.
- **Synthetic mechanism** (multi-hop transitivity with shuffled facts; Fig. 7): the relation is linearly decodable from the frozen residual (0.93–0.97) where the zero-shot decoder sits at chance; a 2-demo budget surfaces the loop's inference but never exceeds the one-pass MLP. The measured limit is **order-locked binding**, not depth.

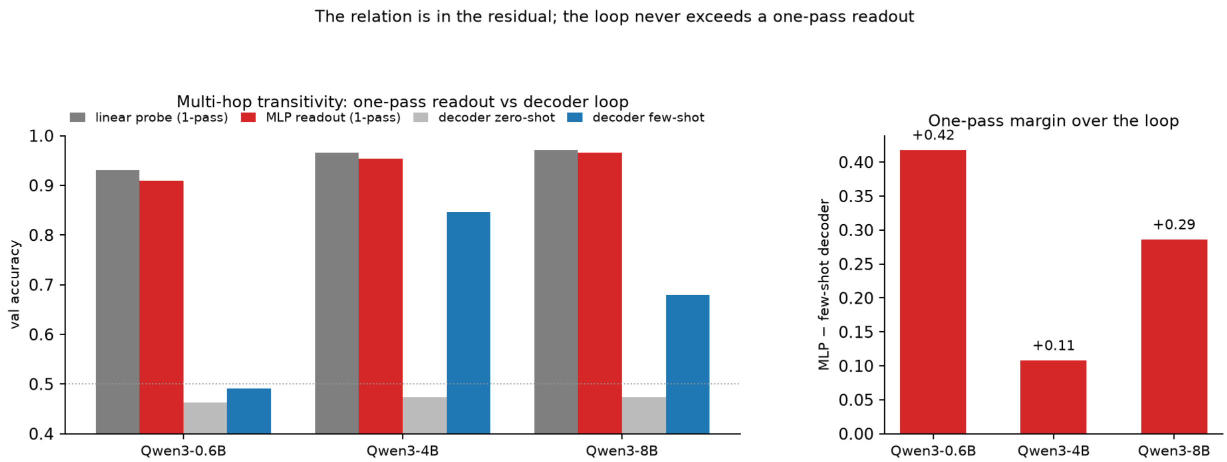


**Figure 7: Mechanism (synthetic).** Multi-hop transitivity: one-pass readout vs decoder loop across Qwen3 scale. The loop never exceeds a one-pass readout.

The remaining "loop wins" cell — a $TC^0$-hard task on which a *frozen* loop decisively beats one-pass — is largely unoccupied in our sweep; the counter-position (COCONUT, looped-latent) buys serial depth by *training*, which is out of scope here. Because RuleTaker is an n2k pilot with thin per-depth bins ($n = 21$–$51$), we treat the serial-depth boundary as *exploratory*: it indicates where a $TC^0$ gap would appear but is underpowered to fix its location.

# 6 Discussion

**The thesis, in its conditional form.** On closed-set, one-pass-representable tasks, a small head on the frozen residual matches a fair autoregressive classifier — so the answer is already in the vector. This is a measurement, not a metaphysics: we do not claim the loop never computes, and we identify the three regimes where it does (§5).

**The residual supports a competitive closed-set classifier.** The readout's largest margins are exactly where the loop is weakest (Granite RuleTaker +12.0pt; 0.6B everywhere) — the readout channel saturates before the verbal channel. This is the operational face of the internal-knowledge/external-behavior disconnect Orgad et al. [24] document and Gekhman et al. [16] formalize as *hidden knowledge*: a producer readout on the prompt residual can match the classifier the loop provides on this task class. That the readout is near-linear (§3) sharpens the point: the answer is a near-linear read of a vector the model already computes.

**The Mistral BoolQ exception.** Mistral-7B is the one model where the loop significantly wins on BoolQ ($-2.2$pt, $p = 2.6\text{e}{-03}$). We report it plainly rather than explain it away: it is a single-model, single-task cell, and Mistral's wide-shallow stack ($32 \times 4096$) plateaus earliest of the campaign (§4.4), so its final-layer residual is the weakest readout substrate relative to its own loop. One cell does not move the map, but it is a real boundary, not noise.

**Separability of readout from generative interface.** The unembedding is one readout of the residual — a fixed, next-token-trained one. That a supervised head matches it is the default expectation once stated that way; the empirical content is *where* it fails (ARC at scale, DSV4 deduction) and that the failure modes are orderly.

**Specialist frontier (acknowledged).** Fine-tuned encoder classifiers (RoBERTa-large $\sim$0.87, DeBERTa-v3 $\sim$0.88, T5-11B 0.91 on BoolQ) **beat** our 8B one-pass readout (0.878). Our claim is readout-vs-loop on a frozen backbone — the comparison a serving stack faces when the weights are already loaded — not readout-vs-fine-tuned-specialist. Where a specialist can be trained and shipped, it wins on raw accuracy; the readout's economics are marginal-cost (the forward pass already happened).

**What we are not claiming.** A frozen continuous latent loop works (it drifts — Exp 7); the loop never computes; we beat fine-tuned specialists; we shipped residual-first inference. Paper 2 explores the serving architecture as a proposal.

# 7 Limitations

- **RuleTaker is an n2k pilot** (2000/1000, natural depth mix, $d3$ fat), not the full test; per-depth bins are thin ($n = 21$–$51$).
- **Multiple testing**: Table 1's 18 McNemar $p$-values are reported uncorrected (stars); all 11 raw-significant cells survive Benjamini–Hochberg FDR at $q{<}0.05$. Placement is stricter: 3 of 9 raw mid-depth wins drop under the same correction (6/18 remain).
- **Truncation is not logged per row**: the readout reads at pad_max $= 384$ (left-truncated) against the loop's 2048–8192, and we do not record the fraction of rows truncated, so a length-asymmetric truncation cost on long-passage rows cannot be ruled out from the logged data (the per-length deconfound shows no accuracy collapse, but it is a coarse check).
- **Supervision asymmetry is bounded, not eliminated**: the head still sees more gold labels than the loop's in-context exemplars (Ext 13 budget-stress-tests the loop at 0.6B–8B BoolQ; Arm C — a fine-tuned decoder, or a probe trained only on the loop's $k$ exemplars — is future work).
- **Reproducibility**: per-layer heads and per-row predictions are persisted for campaign runs; the two oldest headline runs (Ext 8/9) predate head persistence. MoE active-param FLOPs are unmeasured (Table 3).
- **Scope**: closed-set, one-pass-representable tasks only. The loop baseline is a scoring classifier, not chain-of-thought generation. No claim transfers to open-ended decoding.

# 8 Related Work

**Accessing what a frozen model knows** is established from several directions, none of which runs our comparison. *Behavioral* work [27] shows CoT rationales can be unfaithful — text-only access, no representation. *Latent-belief* work reads the representation directly: CCS [5] finds an unsupervised truth direction (belief vs. gold-label answer recovery), pre-CoT belief probes [13], belief-formation timing [3], and the correctness/self-knowledge cluster [6, 16, 21, 24]. Orgad et al. detect errors (and, in their §6, re-rank 30 samples); Gekhman et al. formalize *hidden knowledge* and re-rank up to 1k samples with a verifier probe on $h(q, a)$ — both score *candidates the model already emitted*, not a producer head on the prompt residual alone, and neither holds that head to a fairly-conditioned closed-set loop. *Decoding-suboptimality* work [1, 4, 8, 9] and, closest, Hazenoot et al. [20] show hidden-state readouts beat the token-probability decision rule or zero-shot prompting — baselines their own analyses show are weak. *Probe-for-verification* work [22] uses internal states to improve generation, not replace it.

**Residual-stream lenses and representation tools** ask *what* a layer already encodes, not whether a small gold-label readout matches a fair loop as a classifier. The logit lens [23] and tuned lens [2] decode intermediate residuals toward the vocabulary; Patchscopes [17] and representation engineering [28] intervene on or read directions for interpretability and control. We borrow their premise — the stream is readable mid-network — and measure a different object: gold-answer accuracy of a frozen-residual readout vs. the same model's few-shot scorer.

**The empty cell** — frozen open weights, a producer readout on $h(q)$ fit to gold labels, run head-to-head against a fairly-conditioned closed-set scoring loop under format/shuffle/randproj controls, with supervision budget-stress-tested rather than assumed equal — is the one this paper occupies. Turpin saw behavior, Cox saw belief, Orgad detected errors and re-ranked samples, Gekhman formalized hidden knowledge with a verifier on $h(q, a)$, Cho beat a weak baseline, Hazenoot beat zero-shot prompting, ReProbe verified steps to generate better; we compare a one-pass producer and the loop at producing the right answer, because the weights are open and frozen.

**Delimiting counter-position.** COCONUT [19] and looped-latent transformers [25] *train* a continuous latent loop to buy serial depth — they set the boundary we test without training. Early-exit serving [7, 15, 18, 26] decides *when to exit*; we measure *when the answer is already available* in the residual. The two are complementary: our placement map is evidence for their premise, not a serving design we ship. Trajectory-dynamics probes [14] read layer-wise displacement across all tokens for OOD monitoring — a richer readout than our static one, aimed at a different goal.

# 9 Conclusion

On closed-set, one-pass-representable tasks, a small readout head fit to the frozen residual recovers the answer without generating tokens — and matches the strongest *closed-set scoring* use of the autoregressive loop across the space we measured. Three loop-win *regimes* remain: a frontier MoE on serial deduction (RuleTaker pilot at DeepSeek-V4), Mistral-7B on BoolQ, and a small-model parametric gap (ARC) that closes with scale; elsewhere the loop's edges are statistical ties (four of six BoolQ cells within ±1.2pt). The claim is therefore bounded but strong: on this task class, the vector already has the answer — and we map the boundaries where the loop still earns its keep. The readout is near-linear — the answer is a near-linear read of a vector the model already computes. **The vector already has the answer.** The construction of serving machinery on that substrate is companion work; here, we measured the substrate.

# Code and data availability

Per-layer heads, per-row predictions, run manifests, and the scripts that regenerate Tables 1–2 (`bench.py`, `scripts/build_table1.py`, `scripts/verify_tables.py`) will be released with the arXiv source. Dataset releases used as-is: BoolQ [10] (`google/boolq`), RuleTaker [12] (`tasksource/ruletaker`; n2k pilot subsample), ARC-Challenge [11] (`allenai/ai2_arc`, config `ARC-Challenge`).

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**Table 1: One-pass residual readout vs. fair AR closed-set scorer** (accuracy). Readout: MLP on frozen final residual (*per-seed mean*; identical final-layer numbers as Tables 2 and 4). Loop: next-token log-prob over the closed answer set, full context, balanced few-shot — not free-form generation. loop.0 / loop.8 = zero- / 8-shot; *best loop* = stronger arm ($k$ in parentheses; RT 0.6B and ARC DSV4 prefer zero-shot). $\Delta$ = readout − best loop (best-of makes $\Delta$ conservative). Paired McNemar on identical rows using 4-seed vote preds (mean–vote $\leq$0.9pt; never flips sig/ns). Stars mark *uncorrected* $p$ ($^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$); all 11 raw-significant cells remain significant under Benjamini–Hochberg FDR at $q<0.05$ across the 18 cells. [a]BoolQ val 3270; Qwen $k\leq64$ pad 8192, else $k=8$ pad 2048. [b]RuleTaker n2k pilot, val 1000 (same rows for readout and loop). [c]ARC-Challenge test 1165, $k=8$.

| Task | Model | Readout | loop.0 | loop.8 | best loop | $\Delta$ (readout − loop) | $p$ (McNemar) |
|---|---|---|---|---|---|---|---|
| BoolQ[a] | Qwen3-0.6B | **0.746** | 0.631 | 0.645 | 0.715 (32) | **+0.030**$^{***}$ | 3.2e−05 |
| | Qwen3-4B | 0.856 | 0.855 | 0.865 | **0.867** (64) | −0.012 | 0.094 |
| | Qwen3-8B | 0.878 | 0.862 | 0.876 | **0.886** (16) | −0.008 | 0.243 |
| | Mistral-7B | 0.841 | 0.824 | 0.863 | **0.863** (8) | −0.022$^{**}$ | 0.003 |
| | Granite-3.1-8B | 0.854 | 0.800 | 0.865 | **0.865** (8) | −0.011 | 0.076 |
| | DeepSeek-V4-Flash | 0.896 | 0.888 | 0.906 | **0.906** (8) | −0.009 | 0.078 |
| RuleTaker n2k[b] | Qwen3-0.6B | **0.650** | 0.598 | 0.587 | 0.598 (zero) | **+0.052**$^{**}$ | 0.004 |
| | Qwen3-4B | **0.744** | 0.679 | 0.703 | 0.703 (8) | **+0.041**$^{**}$ | 0.005 |
| | Qwen3-8B | **0.758** | 0.666 | 0.729 | 0.729 (8) | +0.029 | 0.056 |
| | Mistral-7B | **0.643** | 0.517 | 0.571 | 0.571 (8) | **+0.072**$^{***}$ | 8.4e−04 |
| | Granite-3.1-8B | **0.805** | 0.554 | 0.685 | 0.685 (8) | **+0.120**$^{***}$ | 6.5e−12 |
| | DeepSeek-V4-Flash | 0.765 | 0.603 | 0.803 | **0.803** (8) | −0.038$^{**}$ | 0.009 |
| ARC-Challenge[c] | Qwen3-0.6B | 0.492 | 0.502 | 0.597 | **0.597** (8) | −0.106$^{***}$ | 3.2e−11 |
| | Qwen3-4B | 0.841 | 0.826 | 0.882 | **0.882** (8) | −0.041$^{***}$ | 2.9e−06 |
| | Qwen3-8B | 0.909 | 0.902 | 0.913 | **0.913** (8) | −0.004 | 0.567 |
| | Mistral-7B | 0.728 | 0.750 | 0.779 | **0.779** (8) | −0.051$^{***}$ | 8.3e−05 |
| | Granite-3.1-8B | 0.708 | 0.732 | 0.790 | **0.790** (8) | −0.082$^{***}$ | 5.8e−10 |
| | DeepSeek-V4-Flash | 0.951 | 0.961 | 0.959 | **0.961** (zero) | −0.010 | 0.064 |

**Table 2: Readout probe-placement sweeps** (18 cells, all batch 8): best mid-depth tap vs the final-layer readout, paired McNemar on identical rows. *Layers* = stack depth (28/36/36/32/40/43); taps are residual-layer indices. *Plateau onset* = first swept layer within 1 pt of the final-layer readout — the verdict representation is fully assembled from there on; at DeepSeek-V4 this happens by L22/42 (52% depth) on *all three tasks*, with ARC showing the sharpest snap-in (near-chance at L15 $\rightarrow$ 0.944 at L22). Mid-depth taps win significantly at 9/18 cells raw (6/18 after Benjamini–Hochberg FDR at $q<0.05$) and never lose significantly.

| Task | Model | # | Tap | Dep. | Plateau | best | final | $\Delta$ | $p$ |
|---|---|---|---|---|---|---|---|---|---|
| BoolQ | Qwen3-0.6B | 28 | L18 | 64% | L18 (67%) | 0.761 | 0.746 | +0.010 | 0.18 |
| | Qwen3-4B | 36 | L24 | 67% | L24 (69%) | 0.868 | 0.856 | **+0.011**$^{*}$ | 0.038 |
| | Qwen3-8B | 36 | L30 | 83% | L24 (69%) | 0.889 | 0.878 | **+0.011**$^{*}$ | 0.016 |
| | Mistral-7B | 32 | L16 | 50% | **L16 (52%)** | 0.863 | 0.841 | **+0.022**$^{***}$ | 1.1e−04 |
| | Granite-3.1-8B | 40 | L20 | 51% | **L17 (44%)** | 0.868 | 0.854 | **+0.018**$^{**}$ | 1.0e−03 |
| | DeepSeek-V4-Flash | 43 | L36 | 86% | **L22 (52%)** | 0.900 | 0.896 | +0.005 | 0.20 |
| RuleTaker n2k | Qwen3-0.6B | 28 | L18 | 64% | **L13 (48%)** | 0.688 | 0.650 | **+0.036**$^{*}$ | 0.020 |
| | Qwen3-4B | 36 | L24 | 67% | L24 (69%) | 0.781 | 0.744 | **+0.038**$^{**}$ | 2.8e−03 |
| | Qwen3-8B | 36 | L24 | 67% | L24 (69%) | 0.778 | 0.758 | **+0.025**$^{*}$ | 0.040 |
| | Mistral-7B | 32 | L18 | 56% | **L11 (35%)** | 0.702 | 0.643 | **+0.059**$^{***}$ | 5.2e−05 |
| | Granite-3.1-8B | 40 | L20 | 51% | L20 (51%) | 0.816 | 0.805 | +0.011 | 0.37 |
| | DeepSeek-V4-Flash | 43 | L29 | 67% | **L22 (52%)** | 0.776 | 0.765 | +0.021 | 0.057 |
| ARC-Challenge | Qwen3-0.6B | 28 | L26 | 93% | L26 (96%) | 0.500 | 0.492 | +0.013 | 0.082 |
| | Qwen3-4B | 36 | L29 | 81% | L24 (69%) | 0.856 | 0.841 | **+0.022**$^{***}$ | 2.2e−04 |
| | Qwen3-8B | 36 | L35 | 97% | L24 (69%) | 0.909 | 0.909 | +0.001 | 1.00 |
| | Mistral-7B | 32 | L20 | 63% | L18 (58%) | 0.741 | 0.728 | +0.013 | 0.50 |
| | Granite-3.1-8B | 40 | L39 | 100% | L39 (100%) | 0.708 | 0.708 | 0.000 | — |
| | DeepSeek-V4-Flash | 43 | L42 | 98% | **L22 (52%)** | 0.951 | 0.951 | −0.005 | 0.29 |

**Table 3: What each access to the frozen residual costs** (economics orientation, not a benchmark). Status stamps every row: "measured" = our run protocol; "projected" = forward arithmetic we did not run. Readout & scoring-loop are the same single pass at equal context — the readout's structural gap is **no generation**, not layer-skipping. Early-exit serving shares our placement premise and is prior art for any "skip the tail" claim.

| Access | Cost shape | Generates? | Status |
|---|---|---|---|
| readout (final tap) | 1 fwd , short ctx | no | measured |
| scoring loop ".8" | 1 fwd, long ctx | no | measured |
| early-exit decode | 1 fwd, tail truncated | yes | projected / prior art |
| decode / output | $K$ fwds (per token) | yes | measured (no self-stop) |

*Note.* The decode row *is* measured: greedy decode from the Answer: prompt never self-terminates (25/25 hit the 300-token cap at 0.6B; 7/8 hit the 1000-token cap at DeepSeek-V4).

**Table 4: Controls, all 18 cells.** Every headline number clears all four controls. *label-shuffle*: the head refit on permuted labels sits at chance. *random-projection*: a head on random Gaussian projections retains accuracy with permuted-basis $\approx$ max-basis $\gg$ noise floor — the signal is *diffuse*. *full-context*: a mean-pooled full-context readout underperforms the final-token readout — the verdict is computed into the final state, not read off the passage. Balanced chance is 0.50 (BoolQ, RuleTaker) and 0.25 (4-way ARC); the *majority-class* baseline is $\approx$0.62 (BoolQ), 0.50 (RuleTaker), 0.27 (ARC).

| Task | Model | shuffle | rp max | rp perm | rp noise | full-ctx | final |
|---|---|---|---|---|---|---|---|
| BoolQ | Qwen3-0.6B | 0.533 | 0.734 | 0.725 | 0.601 | 0.667 | 0.746 |
| | Qwen3-4B | 0.542 | 0.861 | 0.859 | 0.604 | 0.717 | 0.856 |
| | Qwen3-8B | 0.528 | 0.874 | 0.874 | 0.607 | 0.739 | 0.878 |
| | Mistral-7B | 0.537 | 0.810 | 0.803 | 0.607 | 0.685 | 0.841 |
| | Granite-3.1-8B | 0.532 | 0.848 | 0.849 | 0.607 | 0.679 | 0.854 |
| | DeepSeek-V4 | 0.522 | 0.889 | 0.891 | 0.602 | 0.678 | 0.896 |
| RuleTaker | Qwen3-0.6B | 0.511 | 0.635 | 0.635 | 0.489 | 0.517 | 0.650 |
| | Qwen3-4B | 0.474 | 0.718 | 0.726 | 0.505 | 0.586 | 0.744 |
| | Qwen3-8B | 0.488 | 0.736 | 0.745 | 0.496 | 0.575 | 0.758 |
| | Mistral-7B | 0.495 | 0.645 | 0.634 | 0.496 | 0.503 | 0.643 |
| | Granite-3.1-8B | 0.491 | 0.756 | 0.757 | 0.496 | 0.545 | 0.805 |
| | DeepSeek-V4 | 0.500 | 0.726 | 0.716 | 0.506 | 0.518 | 0.765 |
| ARC | Qwen3-0.6B | 0.278 | 0.470 | 0.470 | 0.254 | 0.278 | 0.492 |
| | Qwen3-4B | 0.269 | 0.802 | 0.803 | 0.255 | 0.349 | 0.841 |
| | Qwen3-8B | 0.287 | 0.869 | 0.867 | 0.251 | 0.399 | 0.909 |
| | Mistral-7B | 0.259 | 0.679 | 0.678 | 0.251 | 0.277 | 0.728 |
| | Granite-3.1-8B | 0.284 | 0.598 | 0.588 | 0.251 | 0.279 | 0.708 |
| | DeepSeek-V4 | 0.269 | 0.942 | 0.940 | 0.236 | 0.439 | 0.951 |

## References


[1] Momin Abbas, Yi Zhou, Parikshit Ram, Nathalie Baracaldo, Horst Samulowitz, Theodoros Salonidis, and Tianyi Chen. Enhancing in-context learning via linear probe calibration. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024.

[2] Nora Belrose, Zach Furman, Logan Smith, Danny Halawi, Igor Ostrovsky, Lev McKinney, Stella Biderman, and Jacob Steinhardt. Eliciting latent predictions from transformers with the tuned lens. *arXiv preprint arXiv:2303.08112*, 2023.

[3] Siddharth Boppana, Annabel Ma, Max Loeffler, Raphael Sarfati, Eric Bigelow, Atticus Geiger, Owen Lewis, and Jack Merullo. Reasoning theater: Disentangling model beliefs from chain-of-thought. *arXiv preprint arXiv:2603.05488*, 2026.

[4] Marcus Buckmann and Edward Hill. Logistic regression makes small LLMs strong and explainable "tens-of-shot" classifiers. *arXiv preprint arXiv:2408.03414*, 2024.

[5] Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. Discovering latent knowledge in language models without supervision. In *International Conference on Learning Representations (ICLR)*, 2023.

[6] Iván Vicente Moreno Cencerrado, Arnau Padrés Masdemont, Anton Gonzalvez Hawthorne, David Demitri Africa, and Lorenzo Pacchiardi. No answer needed: Predicting LLM answer accuracy from question-only linear probes. In *ICLR Workshop on Principled Design for Trustworthy AI*, 2026.

[7] Yanxi Chen, Xuchen Pan, Yaliang Li, Bolin Ding, and Jingren Zhou. EE-LLM: Large-scale training and inference of early-exit large language models with 3D parallelism. In *International Conference on Machine Learning (ICML)*, 2024.

[8] Hakaze Cho, Yoshihiro Sakai, Mariko Kato, Kenshiro Tanaka, Akira Ishii, and Naoya Inoue. Token-based decision criteria are suboptimal in in-context learning. In *Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2025.

[9] Hyunsoo Cho, Hyuhng Joon Kim, Junyeob Kim, Sang-Woo Lee, Sang-goo Lee, Kang Min Yoo, and Taeuk Kim. Prompt-augmented linear probing: Scaling beyond the limit of few-shot in-context learners. In *AAAI Conference on Artificial Intelligence*, 2023.

[10] Christopher Clark, Kenton Lee, Ming-Wei Chang, Tom Kwiatkowski, Michael Collins, and Kristina Toutanova. BoolQ: Exploring the surprising difficulty of natural yes/no questions. In *Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2019.

[11] Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? try ARC, the AI2 reasoning challenge. *arXiv preprint arXiv:1803.05457*, 2018.

[12] Peter Clark, Oyvind Tafjord, and Kyle Richardson. Transformers as soft reasoners over language. In *International Joint Conference on Artificial Intelligence (IJCAI)*, 2020.

[13] Kyle Cox, Darius Kianersi, and Adrià Garriga-Alonso. Post-hoc reasoning in chain of thought: Decoding and steering pre-committed answers. *arXiv preprint arXiv:2603.01437*, 2026.

[14] Hamed Damirchi, Ignacio Meza De la Jara, Ehsan Abbasnejad, Afshar Shamsi, Zhen Zhang, and Javen Shi. Truth as a trajectory: What internal representations reveal about large language model reasoning. *arXiv preprint arXiv:2603.01326*, 2026.

[15] Mostafa Elhoushi, Akshat Shrivastava, Diana Liskovich, Basil Hosmer, Bram Wasti, Liangzhen Lai, Anas Mahmoud, Bilge Acun, Saurabh Agarwal, Ahmed Roman, Ahmed A Aly, Beidi Chen, and Carole-Jean Wu. LayerSkip: Enabling early exit inference and self-speculative decoding. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.

[16] Zorik Gekhman, Eyal Ben David, Hadas Orgad, Eran Ofek, Yonatan Belinkov, Idan Szpektor, Jonathan Herzig, and Roi Reichart. Inside-out: Hidden factual knowledge in LLMs. In *Conference on Language Modeling (COLM)*, 2025.

[17] Asma Ghandeharioun, Avi Caciularu, Adam Pearce, Lucas Dixon, and Mor Geva. Patchscopes: A unifying framework for inspecting hidden representations of language models. In *International Conference on Machine Learning (ICML)*, 2024.

[18] Andrey Gromov, Kushal Tirumala, Hassan Shapourian, Paolo Glorioso, and Daniel A. Roberts. The unreasonable ineffectiveness of the deeper layers. In *International Conference on Learning Representations (ICLR)*, 2025.

[19] Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian. Training large language models to reason in a continuous latent space. In *Conference on Language Modeling (COLM)*, 2025.

[20] Luc Hazenoot, Zhaochun Ren, and Amirhossein Zohrehvand. Measuring concept content in text from LLM activations: ESG evidence from concept vectors and linear probes. *arXiv preprint arXiv:2608.07208*, 2026.

[21] Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, Scott Johnston, Sheer El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Bai, Sam Bowman, Stanislav Fort, Deep Ganguli, Danny Hernandez, Josh Jacobson, Jackson Kernion, Shauna Kravec, Liane Lovitt, Kamal Ndousse, Catherine Olsson, Sam Ringer, Dario Amodei, Tom Brown, Jack Clark, Nicholas Joseph, Ben Mann, Sam McCandlish, Chris Olah, and Jared Kaplan. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

[22] Jingwei Ni, Ekaterina Fadeeva, Tianyi Wu, Mubashara Akhtar, Jiaheng Zhang, Elliott Ash, Markus Leippold, Timothy Baldwin, See-Kiong Ng, Artem Shelmanov, and Mrinmaya Sachan. ReProbe: Efficient test-time scaling of multi-step reasoning by probing internal states of large language models. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026.

[23] nostalgebraist. Interpreting GPT: The logit lens. LessWrong, 2020. URL https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens.

[24] Hadas Orgad, Michael Toker, Zorik Gekhman, Roi Reichart, Idan Szpektor, Hadas Kotek, and Yonatan Belinkov. LLMs know more than they show: On the intrinsic representation of LLM hallucinations. In *International Conference on Learning Representations (ICLR)*, 2025.

[25] Nikunj Saunshi, Nishanth Dikkala, Zhiyuan Li, Sanjiv Kumar, and Sashank J. Reddi. Reasoning with latent thoughts: On the power of looped transformers. In *International Conference on Learning Representations (ICLR)*, 2025.

[26] Tal Schuster, Adam Fisch, Jai Gupta, Mostafa Dehghani, Dara Bahri, Vinh Q. Tran, Yi Tay, and Donald Metzler. Confident adaptive language modeling. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[27] Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[28] Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.